

Frequency or Information? A Controlled Study of Spectral Side-Channel Conditioning in Graph Diffusion

Tamara Bluzer Dan Shamia Daniel Halperin Itamar Kolodny

315287441 208004119 207826314 211490362

Tel Aviv University

Abstract

Spectral conditioning is standard in graph generation, but in existing generators the truncated spectrum *is* the representation being generated, which ties “which frequency band helps” to “how much of the graph that representation already carries.” We separate the two: we diffuse the full adjacency matrix and supply the truncated normalized-Laplacian spectrum only as a *side channel*, so band and cutoff k are the only quantities that vary across arms, and we add a model-free probe measuring how much of the target each condition gives away. This is an exploratory study — one benchmark, one protocol — and the headline arms are *oracle-conditioned*: the spectra are taken from the real evaluation graphs, so those arms are a diagnostic upper bound rather than a generative model. Replacing the oracle with a learned spectral prior costs most of the benefit (Ratio $5.92 \rightarrow 159.72$ for `high` $k=32$, against an Erdős–Rényi reference of 161.67, with `low` $k=8$ failing its gate outright). On SPECTRE’s Planar benchmark, a sweep of five conditioning arms over five cutoffs (one run per cell) shows no consistent monotonic trend across the tested cutoffs: 23 of 25 cells sit inside the spread of the unconditioned baseline, and the effect is confined to two isolated minima — and the probe finds that both of them leak. `low` $k=8$ improves the MMD Ratio $2.3\times$ while recovering 72.7% of the target’s edges; `high` $k=32$ improves it $29.7\times$ while recovering 98.5%, near enough to a near-oracle regime to be read as one. In the one arm whose generated graphs survive, those graphs retrieve their own conditioning graph first in all 32 cases (edge F_1 0.65 against 0.09 for mismatched pairs), so the advantage is at least partly reconstruction of the evaluation set. We therefore cannot attribute either minimum to frequency content. What we can still exclude is recovery as a *predictor* of quality: it rises smoothly with k in both bands while quality stays flat with two isolated spikes, so it does

not locate the minima it now implicates.

1 Introduction

Topology-only graph generation learns a distribution over adjacency matrices with no node or edge features, so connectivity is the only thing modelled. Diffusion models now dominate the area (Niu et al., 2020; Jo et al., 2022; Vignac et al., 2023), and a recurring idea is to hand the generator the graph’s Laplacian spectrum. Martinkus et al. (2022) generate eigenpairs with a GAN and decode an adjacency matrix from them; Minello et al. (2025) diffuse the spectrum itself; Vignac et al. (2023) feed Laplacian eigenfeatures to their denoiser as fixed extra inputs; and Zhu et al. (2025) argue, for representation learning, that the low-frequency part of the spectrum carries the useful global information.

These motivate an obvious question — *which* part of the spectrum helps generation? — that existing designs cannot answer cleanly. When the truncated spectrum (λ_k, U_k) is the representation the model generates and decodes from, k controls the frequency content of the condition *and*, simultaneously, how much of the target that representation determines. On $n=64$ planar graphs $U_k \text{diag}(\lambda_k) U_k^\top$ at $k=32$ is close to the Laplacian itself, so a band that “wins” there may simply be inverting its own condition. No cited work reports a condition-information floor against which its frequency curve could be read.

Research question. *Given a diffusion model whose state already carries the full graph, does an explicit spectral side channel improve generation, and does the incremental benefit depend on which frequency band and how many eigenpairs are supplied?*

Phrasing it as a side channel makes the question falsifiable. Every arm diffuses the same object with the same architecture and budget and

receives a condition tensor of the same shape; only the *selection* of eigenpairs differs. A `none` arm gives the null, and three controls — a random band, shape-matched Gaussian noise, and another graph’s real low band — separate frequency-specific information from conditioning capacity and from generic spectral statistics. We then add the measurement the literature is missing: a model-free probe that reconstructs edges from the condition alone.¹

Our contributions are that design; the central negative result that there is **no consistent monotonic trend across the tested cutoffs** (§4.2); a leakage probe that **overturned our own headline claim** (§4.3) — an earlier version of this analysis reported the winning band as leak-free, and the corrected probe finds it the highest-leaking arm at its own cutoff, so the confound we claimed to have ruled out stands; a diagnosis of the universal 0% validity (§4.4); and two further negative results (§§4.5–4.6). Throughout we report differences observed under one protocol on one benchmark; we locate an effect, we do not establish the mechanism behind it.

2 Related Work

SPECTRE (Martinkus et al., 2022) is the closest methodological and evaluation reference: a GAN generates (λ_k, U_k) on a Stiefel manifold, then a graph network builds the adjacency matrix from $U_k \text{diag}(\lambda_k) U_k^\top$. It already sweeps $k \in \{2, 4, 8, 16, 32\}$ on Planar and SBM — the grid we reuse — but picks one k per dataset heuristically and never reports the curve. We treat k and the band as the object of study, replace its sign-canonicalization heuristic with a sign- and basis-invariant pair channel, and diffuse adjacency rather than decoding a generated spectrum. We also offset the band by the component count, where SPECTRE drops a fixed `eigvals[1:]`, so a disconnected graph does not waste band on component indicators.

GGSD (Minello et al., 2025) already measures smallest- versus largest-eigenpair representations inside a diffusion generator, so frequency-band sensitivity is not itself novel. Our contribution is

¹Code, run artifacts and the figure-regeneration script: https://github.com/TamaraBluzer/Frequency_aware_diffusion-. Running `scripts/make_paper_figures.py` rebuilds the figure from `results/*.json`; it contains no hand-typed number.

narrower: the incremental value of a side channel while the diffusion state still carries the full graph, with matched-capacity controls and a measured information floor.

Others. **DiGress** (Vignac et al., 2023) supplies our datasets, discrete diffusion process and evaluation suite; it feeds Laplacian eigenvalues to its denoiser as a fixed input augmentation, never as a variable to ablate, and we isolate exactly that variable. **LGD** (Zhou et al., 2024) runs latent diffusion on molecules, never on Planar or SBM, conditioning on scalar properties rather than spectra; §3 says why we abandoned its design. **DualDiff** (Xie and Pan, 2026) couples node- and cluster-level latents, but a cluster assignment collapses global structure into one number K , whereas an eigenpair-indexed condition exposes an ordered band — which is what makes a sweep possible. **SDMG** (Zhu et al., 2025) shows low-frequency reconstruction beating full-spectrum reconstruction for *representation* learning; we ask whether that transfers to *generation*, and it does, but only in one narrow cell and for a reason its framing does not predict.

3 Method

Data. SPECTRE’s released Planar benchmark: 200 graphs at $n=64$, each a Delaunay triangulation of uniform random points, split 128/32/40 following DiGress’s recipe, which we re-process ourselves since its `process()` appends every graph twice. Each arm generates as many graphs as the evaluation set holds (32). **Every number in this paper — every Ratio, every validity fraction, every validation loss — is scored on the validation split**, and that same split selected the checkpoints (since lost). No test-split number is reported anywhere. All results are therefore optimistic in the usual way, and none of them should be read as held-out performance.

Spectral condition. Let $L = I - D^{-1/2} A D^{-1/2} = U \Lambda U^\top$ with $0 = \lambda_1 \leq \dots \leq \lambda_n \leq 2$, and let t be the number of zero eigenvalues (equivalently, connected components). The arms take k eigenpairs as `low` = $(\lambda_{t+1} \dots \lambda_{t+k})$, `high` = $(\lambda_{n-k+1} \dots \lambda_n)$, `random` = k indices drawn uniformly from $t+1 \dots n$, `gaussian` = i.i.d. $\mathcal{N}(0, 1)$ tensors of identical shape, `shuffled` = another graph’s real low band under a derangement, and `none` =

an all-zero condition with a presence flag.

The condition enters the denoiser twice: as a pair channel $(U_k \text{diag}(\lambda_k) U_k^\top)_{ij}$, rescaled per graph to unit RMS so low and high cannot differ in activation scale, and as the selected eigenvalues mapped from $[0, 2]$ to $[-1, 1]$, supplied globally with the timestep. The pair channel is invariant to eigenvector sign flips and to rotations within an exactly repeated eigenspace, sidestepping the ambiguity that motivates SignNet (Lim et al., 2023) without depending on it.

Model. The diffusion state is the centred adjacency matrix $x_0 = 2A - 1$. A cosine forward process (Nichol and Dhariwal, 2021) adds symmetric Gaussian noise to each undirected pair, and a permutation-equivariant DiGress transformer predicts x_0 from x_t (Ho et al., 2020; Vignac et al., 2023), masking the diagonal and padding at every step. It has 4 layers ($d_x=128$, $d_e=64$, $d_y=128$, 8 heads; 1.83M parameters), $T=100$ steps, and trains 200 epochs at batch 16, learning rate 3×10^{-4} , with the condition dropped at probability 0.1 so guidance scale is available at sampling time (Ho and Salimans, 2022). We began from an LGD-style autoencoder and abandoned it after measuring it: its per-pair latent is an expansion, not a compression (32,768 floats for 2,016 binary decisions), and hits $F1 = 1.000$ only by copying A_{ij} down a private lane.

Deviations from our project proposal. Two, both deliberate, both worth stating because they change what the study measures. (i) The proposal specified *latent* diffusion, an $\epsilon_\theta(z_t, t, C_k)$ denoiser over an LGD-style autoencoder’s latent. We diffuse the adjacency matrix instead, for the reason just measured: the autoencoder we built is an expansion rather than a compression and reaches perfect reconstruction by copying A_{ij} , so any latent result would have been a result about that shortcut. Adjacency diffusion also sharpens the side-channel question, since the diffusion state then carries the full graph in every arm. The planned latent-transfer ablation, which would have tested whether the finding survives a genuinely linear-size latent, was not run. (ii) The proposal’s downstream task was planar-versus-SBM classification; we replaced it with SBM community-count prediction (§4.6), because planar-versus-SBM is separable by mean degree alone and so cannot distinguish one augmentation source from another.

Evaluation. We reimplement the GraphRNN/GRAN/SPECTRE suite (You et al., 2018; Liao et al., 2019; Martinkus et al., 2022): MMD (Gretton et al., 2012) under a total-variation Gaussian kernel over degree histograms, clustering coefficients, normalized-Laplacian eigenvalues, 12 ab-spline wavelet scales, and 4-node orbit counts via ORCA (Hočevár and Demšar, 2014). Raw MMD is uninterpretable, so we report **Ratio**: each metric divided by the same metric between two real samples (the train-vs-test floor), averaged, so 1.0 means indistinguishable from real data at this sample size. Validity is networkx planarity and connectivity (Hagberg et al., 2008). We calibrated the harness before training any model of ours: real-vs-real scores ≈ 0 , the floor is 1.00 by construction, and a *per-graph density-matched* Erdős–Rényi control (Erdős and Rényi, 1959) scores $3035 \times$ it.

Protocol and seed coverage. Seed counts differ by experiment and every table states its own. The $k=8$ comparison (Table 1) is three seeds per arm at equal compute; the 5×5 sweep (§4.2) is *a single run per cell, at seed 0*; the re-run of the two minima (Table 2) is three seeds per arm; the downstream classifier (§4.6) is five seeds per arm; the end-to-end and discrete-diffusion runs of §4.5 are single runs. Where three seeds exist we report a paired hierarchical bootstrap (1,000 resamples) over graphs and seeds. It resamples the two levels that actually vary inside a run, so its intervals describe how stable that particular comparison is; they are not a test over conditioning arms, cutoffs or datasets in general.

Leakage probe. The **leakage probe** is model-free: it scores every node pair by $(U_k \text{diag}(\lambda_k) U_k^\top)_{ij}$ — exactly the tensor the denoiser receives — and keeps the m highest-scoring pairs, where m is the target’s true edge count, under whichever of the two global sign orientations scores better. Those true counts are deliberately privileged, but the probe is one decoder, not a bound over decoders: a *high* score is conclusive — the band demonstrably hands the answer over — while a *low* score bounds only this rank-and-threshold family, since a band could encode edges in a form the rule cannot read. Gaussian conditions score at the noise floor, and the full spectrum recovers 100.0%, as it must.

band	Ratio ↓	val. loss ↓	valid
none	327.2 ± 8.8	0.132 ± 0.012	0%
low	158.0 ± 16.2	0.053 ± 0.004	0%
high	339.1 ± 9.2	0.094 ± 0.009	0%
random	326.5 ± 11.8	0.128 ± 0.012	0%

Table 1: Oracle conditioning at $k=8$ — every sample conditioned on a real validation graph’s own spectrum — scored on the validation split that also selected the checkpoints. Three seeds per arm, mean $\pm$ std over seeds. Validation loss is metric-free, and it agrees with Ratio only on low being best: Ratio orders low < random < none < high, validation loss orders low < high < random < none. high is thus the worst arm on Ratio and the second best on loss.

4 Results

Two labels apply to everything below. Every number is scored on the Planar *validation* split, the same split that selected the checkpoints. And the arms of Tables 1 and 2 and of Figure 1 are *oracle-conditioned*: each sample is conditioned on the spectrum of a real validation graph, which makes them a diagnostic upper bound rather than a generative model. §4.5 removes the oracle and reports what that costs.

4.1 Conditioning helps at $k=8$

Table 1 is the kill gate we set before anything downstream, and the first place numbers appear: three seeds per arm, oracle spectra, validation split. Low-frequency conditioning halves the Ratio and cuts validation loss by 60%, while high and random conditioning land where none lands. Paired hierarchical bootstrap 95% intervals for low minus none, high and random are $[-189.4, -152.1]$, $[-206.7, -152.4]$ and $[-195.1, -142.2]$, none of them containing zero. That resampling runs over graphs and seeds within this protocol, so the intervals say how stable this particular comparison is; they do not license a claim about conditioning in general. The model demonstrably uses the condition: under identical diffusion noise, adding it changes 12.95% of edge decisions.

4.2 The sweep: two isolated minima, no monotonic trend

SDMG’s intuition implies quality should vary smoothly with how much of which end of the spectrum is supplied. In this sweep it does not (Figure 1(a)). The 23 conditioned cells that are not one of the two minima span 270.5 to 347.9, with

arm	Ratio ↓	vs. base	leak
high $k=32$	11.1 ± 0.6	29.7×	98.5%
low $k=8$	141.9 ± 19.3	2.3×	72.7%
none	330.1 ± 6.2	—	—

Table 2: Three-seed confirmation of the two minima, with edge recovery from the condition alone. Both reproduce, and both leak: low at $k=8$ recovers far above the 19.2% a random band of the same size recovers at the same cutoff.

mean 327.2 and a spread of 18.5, and the unconditioned baseline of 335.7 falls inside that range: the flat region simply is the baseline. The two minima fall far outside it — 116.3 for low, $k=8$ and 10.5 for high, $k=32$, against a best-of-the-rest of 270.5 — while their neighbours in the same band show nothing: low is 316.4 at $k=4$ and 270.5 at $k=16$. Two cautions on how to read that spread. Each cell is a single run at seed 0, so the sweep is a description of 25 configurations rather than an estimate of any one of them; and the 18.5 is the spread *across different configurations*, not a sampling distribution, so we quote distances from it in Ratio points and never as standard deviations or significance.

Both minima reproduce across three seeds (Table 2): high, $k=32$ is very stable (std 0.58 on a mean of 11.11) and the only arm in the sweep that produced a valid planar graph. low, $k=8$ is noisier but never approaches baseline; its seed-0 draw of 116.27 was its most favourable, so 141.91 is the number to quote. The controls do their job: gaussian spans 322–333 and shuffled — another graph’s real low band — spans 329–342, both flat on the baseline at every k . Neither *having a condition* nor *having real spectral statistics* suffices; the effect needs the matching graph’s own spectrum at a particular band and cutoff, which is the cleanest evidence we have *against* an extra-capacity explanation. It does not tell us what the matching spectrum supplies at those two cells, and we do not claim to know.

4.3 Both minima leak, and leakage does not locate them

An obvious objection to Table 1: each sample is conditioned on a validation graph’s spectrum and scored against those same graphs, so a band that hands over more of the target wins for reasons unrelated to frequency. Table 3 measures it, and the measurement goes against us. At $k=8$ the order-

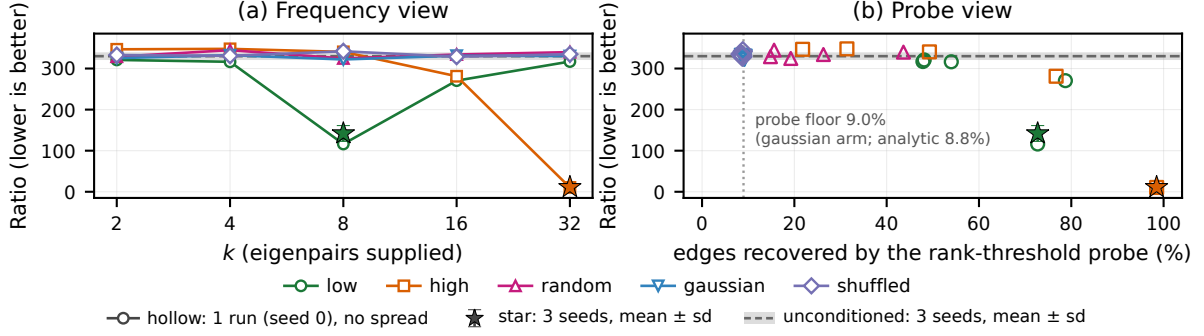


Figure 1: Five conditioning arms $\times$ five cutoffs on Planar, one run per cell at seed 0 on the validation split; stars mark three-seed means $\pm$ std for the two re-run cells, and colours are shared across panels. (a) Quality against k , unconditioned baseline dashed. (b) The same points against measured leakage.

band	$k=2$	$k=4$	$k=8$	$k=16$	$k=32$
low	48.1	54.0	72.7	78.7	47.9
high	21.8	31.4	49.3	76.7	98.5
random	14.8	15.6	19.2	26.3	43.6
gaussian	9.2	8.5	9.2	8.7	9.4
shuffled	8.7	8.7	8.8	8.7	8.3

Table 3: Edge recovery (%) from the condition alone, Planar validation split; chance is 8.8%. Bold marks the two cells that minimise Ratio: both leak, and at $k=8$ the better-generating band is the higher-leaking one. gaussian and shuffled sit at the chance rate at every k , so the controls still do their job. The probe is privileged — it is handed the true node and edge counts — but it is a single decoder, so a high score is conclusive while a low score bounds only this rank-and-threshold family.

ing is low 72.7% > high 49.3% > random 19.2%, with both noise controls at chance: the best-performing arm is also the highest-leaking one. An earlier version of this analysis reported low at 0.0% and concluded that leakage was *anti*-correlated with quality; that was an artifact of a defect in the probe (§5), and we withdraw it. The objection stands as a live confound, and nothing we ran excludes it.

What Table 3 does exclude is leakage as a *locator* of the two minima (Figure 1(b)): recovery climbs smoothly with k in both bands, while quality is flat with two isolated spikes. Low-band recovery peaks not at the minimum but at $k=16$, which recovers more (78.7% against 72.7%) and generates 2.3 $\times$ worse (270.5 against 116.3); in the high band, 76.7% recovery at $k=16$ sits at Ratio 281.2, inside the flat region, while 98.5% at $k=32$ collapses to 10.5. Recovery is therefore not sufficient to predict quality, and reading the $k=8$ min-

imum purely as leakage does not fit the shape of the data. That is the honest residual: a confound we can no longer exclude, which nevertheless fails to explain where the minima are. This defence is Planar-only: it rests on the non-monotone low-band peak at $k=16$, and we have no SBM generation sweep to pair against the SBM leakage grid.

The two minima do still differ in degree. At 98.5% recovery high, $k=32$ is close to being handed the target outright, so its 29.7 $\times$ improvement should not be reported as a frequency effect on any reading; low, $k=8$ leaks substantially less, but less is not none. What replicates on SBM is the leakage itself: both bands run far above the 9.4% chance level, the low band rising monotonically in k (33.8% at $k=2$ to 74.4% at $k=32$) against high’s 23.0% to 87.7%. So the confound is not an artifact of Planar’s construction from 2D point sets — but the Planar low band’s non-monotone shape is not reproduced there, which is the second reason the paragraph above is a one-dataset argument.

The samples themselves reconstruct their conditioning graph. The probe bounds what the condition *carries*; it does not say what the model *did* with it. For the one arm whose generated graphs we still hold — low, $k=8$, seed 0 — we can ask directly, by comparing each sample against the validation graph whose spectrum conditioned it. The answer is unambiguous. Matched pairs share edge $F_1 = 0.653$ against 0.088 for mismatched pairs (Jaccard 0.485 against 0.046; 115.4 of ~ 177 edges in common against 15.6), a gap of 0.565 with a bootstrap 95% interval of [0.556, 0.574] over the 32 graphs. Scoring every sample against every candidate retrieves its own

conditioning graph first *in all 32 cases*, mean rank 1.0 of 32 against a chance rate of 3.1%. Each sample is individually identifiable as a partial copy of the specific held-out graph it was conditioned on, and that graph is in the reference set the sample is then scored against. This is direct evidence for the confound, not merely an inability to exclude it: the arm’s MMD advantage is at least partly reconstruction of the evaluation data. One measurement points the other way — a sample’s own low band is no closer to its conditioning spectrum than to any other graph’s (retrieval at 3.1%, exactly chance) — so the copying is of edges, not of the conditioned quantity, and we cannot say by what route. We stress the coverage limit: this is 1 arm of 12, the remaining sample files are lost (§5), and without them we cannot ask whether $k=16$, which recovers more yet generates worse, reconstructs less. That comparison is the one that would settle the section, and we cannot run it.

4.4 Why validity is zero everywhere

Validity is *connected AND planar*, so the universal 0% cannot say which half failed. We separate them two ways. On the generated graphs themselves — again only `low`, $k=8$, seed 0, the one saved arm — 46.9% are connected (bootstrap 95% interval [31.3%, 62.5%] over 32 graphs, mean 1.66 components) and 0.0% are planar, so planarity is the binding constraint and connectivity is roughly a coin flip. Only 9.4% exceed the $3n - 6$ edge bound, so these are not merely too dense: they are the right size and the wrong shape. The probe’s reconstructions separate the two halves the same way across the whole grid. Connectivity is easy, and cheap in information: by $k=32$ every reconstruction is connected in both bands, and even *shuffled* — another graph’s low band, at the noise floor — is 71.9% connected at $k=16$ and 100% at $k=32$. Planarity essentially never is. Near-complete recovery does not buy it: `low` at $k=16$ recovers 78.7% of the target’s edges and `high` at $k=16$ recovers 76.7%, and neither yields a single planar graph. Only `high`, $k=32$ reaches planarity at all, at 3.1%.

That arm explains why. At 98.5% recovery the mean number of misplaced edges is 2.7 of ~ 177 (max 6), and the fraction with *zero* misplaced edges is 3.1% — the same 3.1% that is planar, so every reconstruction with even one wrong edge is non-planar. This is a property of the target class, not of our model: a planar graph on $n=64$ with

177 edges sits near the $3n - 6 = 186$ bound, so it is nearly maximally triangulated and almost any added chord creates a K_5 or $K_{3,3}$ minor. Validity here is therefore close to a binary test of exactness rather than a graded quality measure, which is why an arm can improve Ratio $2.3\times$ while pinned at zero. The usual prescription — add structural features — has negative precedent in our own runs: the discrete arm of §4.5 includes cycle counts and still reports 0% validity at every scale. We had hoped a graded *distance-to-planarity* would separate arms the boolean cannot, and report here that it does not. The measurement is greedy — repeatedly delete an edge of a reported Kuratowski subgraph, up to a cap of 12 — so it is an upper bound on the deletions a graph needs, never the minimum. At that cap it is also almost entirely right-censored: on Planar, *every* graph in 24 of the 25 cells exceeds it, and the single exception, `high` $k=32$, terminates on 3 of 32 graphs, at distances 0, 4 and 11. On SBM no cell has a single terminating graph. A median over three survivors is not a property of an arm, and the medians we previously quoted were exactly that, so we withdraw them. The honest summary is that these reconstructions are more than 12 edge deletions from planar essentially everywhere, which is a bound rather than a ranking; separating arms this way would need a far larger cap and an exact solver we do not have.

4.5 From oracle to end-to-end

Everything above is oracle-conditioned: the spectra come from held-out real graphs, making these arms a diagnostic upper bound rather than a generative model — SPECTRE’s “real spectra” distinction. To close it we trained a diffusion prior over (n, λ_k, U_k) with an orthogonality penalty and a QR retraction at the final step, then sampled end to end with no ground-truth input. Against an ER reference of 161.67, oracle spectra give Ratio 5.92 (`high`, $k=32$) and 74.86 (`low`, $k=8$); end to end at guidance scale 1 these become 159.72 and 295.64, so the low arm fails its gate outright. Guidance is U-shaped over $\{0.5, 1, 1.5, 2, 3\}$ with an optimum at 1.5, and even there a $24\times$ degradation from oracle remains, because small eigenvector errors compound through $U \text{diag}(\lambda) U^\top$. These Ratios omit ORCA and are *not* comparable to the tables above, which is why the ER reference is quoted beside them. Every configuration in this subsection is a single run, scored on the same validation split.

A discrete-diffusion arm (Austin et al., 2021) sharpens the caveat. At a serious budget (10 layers, 500 steps, 1,000 epochs) the *unconditioned* baseline improves $7.7\times$ over the same process at pilot scale, from 306.78 to 39.60; conditioning still helps, to 15.40 (-61%), but the gap narrows from 276.7 Ratio points to 24.2. (Earlier drafts quoted ~ 327 here, which is the *continuous* baseline of Table 1, not this process’s own pilot.) A headline from a small budget overstates conditioning’s contribution, and validity stays 0% at every scale.

4.6 Downstream utility: a null result, and a labelling flaw that limits it

The intended payoff of better topology generation is synthetic augmentation. We test it on a task deliberately aligned with the hypothesis — predicting an SBM graph’s community count (2–5), which is what low-frequency eigenvalues encode — with 20/40/80 real training graphs, five seeds per arm, and one augmented graph added per real training graph. Diffusion augmentation never beats real-only training: at $n=20$ unconditioned augmentation scores 89.8% against 94.8% real-only and conditioned augmentation 92.0%; at $n=80$ both stay below the 97.2% baseline. Fresh samples from the *true* SBM generator, our control arm, do help slightly (98.8% at $n=80$), so the classifier pipeline itself works.

What these arms can and cannot show. Re-reading `scripts/downstream_classifier.py` for this write-up, we found that the labelling procedure does not support the comparison we first drew from it. Generated graphs are never labelled from their own structure: each one inherits, positionally, the community count of the real template it was sampled alongside. For the conditioned arm that is at least defensible, since the template’s spectrum is the condition. For the unconditioned arm it is not. That arm samples with `band=None`, an all-zero condition, so the only thing it takes from its template is the node count; the generated graph is statistically independent of the label it is handed. At the 1:1 augmentation ratio used here, half of that arm’s training set is graphs whose labels carry no information about them, and that alone is enough to produce the drop we observed. The result is a fact about label noise, not about unconditioned sample quality, and we withdraw the latter reading.

The conditioned arm is not a clean contrast either. It was run at *high*, $k=32$ — the high-leakage extreme of Table 3 — and conditioned on the real training graphs themselves, so its samples are nearer to near-duplicates of the training set than to new augmentation, and its inherited labels are close to correct for exactly the reason that makes the samples redundant. The true-SBM control is the only arm whose labels are produced together with the graph. What survives is narrow: on this task, at these sizes, under this labelling, no diffusion arm beat real-only training, and the task is easy enough to begin with (94.8% from 20 real graphs) that there is little headroom for augmentation of any kind. A usable version of the experiment needs labels recovered from each generated graph — an estimated block count, say — rather than inherited from a template; we did not run that, so this section does not tell us whether the improvements of §4.2 can become downstream utility.

5 Discussion and Limitations

What the side channel buys. Very little, almost everywhere: given a diffusion state that already carries the full graph, a truncated spectrum changed nothing we could measure in 23 of the 25 configurations we ran, and the two exceptions differ in kind. The observation we would defend is *low*, $k=8$ — a $2.3\times$ Ratio improvement and 60% lower validation loss under oracle conditioning on the validation split — and we would defend it only as an observation, because the same condition gives a privileged probe 72.7% of the target’s edges, so a frequency effect and an information effect are not separated here. Coarse global structure the model cannot otherwise infer was our guess at what that condition supplies; leakage is now an equally live one, nothing we ran tests either, and the observation itself rests on three seeds at one cutoff on one dataset. *high*, $k=32$ is the better number and the weaker result: without the probe we would have reported a near-oracle regime as our headline. That a controlled study can invert its own top-line claim — twice: once when the probe was added, and again when the probe was fixed — is our best argument for the information floor none of the cited work reports.

Two caveats on our own reporting. First, the sharpness is unexplained: both minima sit far below every other cell we measured, and both re-

produce across three seeds, so neither is a one-off draw — but a mechanism predicting *which* cells drop is missing. Second, Ratio averages whichever metrics were available at run time, and orbit’s tiny floor dominates any mean it enters: none is 321.21 with ORCA and 125.25 without — a $2.56\times$ gap that explains an ER reference shift (337.91 vs. 161.67) we had treated as unexplained. Decomposing the $k=8$ improvement, orbit alone contributes 84.0% of the -52.9% aggregate while *degree moves the wrong way*; a geometric mean puts it at -34.8% . The low-band result survives — it wins on four of five metrics and on validation loss — but the aggregate overstates it.

Limitations. (i) Every number is scored on the validation split, which also selected the checkpoints, so nothing reported here is held-out performance; seed coverage is uneven (three seeds for Tables 1 and 2, five for §4.6, a single run for each cell of the sweep and for every configuration in §4.5); and the quality sweep is Planar only — on SBM we ran the leakage half, whose corrected pattern replicates, but not the generation half. (ii) The sweep’s values were transcribed from a recycled Colab session’s stdout, so per-metric breakdowns for those runs are lost (ORCA was verified present, so they are five-metric). (iii) Validity is 0% throughout, so we compare distributional fidelity, not sample correctness, and our reduced discrete model is not a DiGress reproduction. (iv) SBM validity uses a spectral-clustering stand-in. (v) Four planned ablations — cluster conditioning, a combinatorial Laplacian, SignNet, and a memorization audit — were not run. (vi) The leakage probe itself was defective when this analysis was first written: its per-graph orientation selection never ran, so every previously reported figure was measured under a single fixed sign, and the low band was reported as leak-free at 0.0% recovery where it in fact recovers 72.7% at $k=8$. Table 3 reports the corrected grid; the frozen prefix numbers are retained in the repository alongside it, as `condition_leakage*_single_orientation.json`. A second defect is uncorrected and is larger than we first thought: the probe sizes its reconstruction from the donor’s node count where the model’s conditioning path crops to the target’s, so on SBM, where n ranges over 55–172, all 32 of 32 shuffled reconstructions have the wrong node count — by up to 108 nodes. That row’s elevation above chance is an

artifact of the probe rather than a property of the condition, so the `shuffled` control is weaker on SBM than even our earlier caveat allowed and we draw nothing from it. Planar’s fixed $n=64$ makes the defect a no-op there. Together these make this an exploratory study: it reports differences observed under one protocol on one benchmark, and leaves the mechanism behind them open.

6 Conclusion

This is an exploratory study of one benchmark under one protocol, with oracle spectra for its headline arms, and what it can support is correspondingly narrow. Measured as a controlled side channel with matched-capacity controls and an information floor: no consistent monotonic trend across the tested cutoffs, two isolated minima that both leak, and an information floor that — once it was fixed — refuted our own claim to have ruled leakage out, while still failing to explain where those minima are.

References

- Jacob Austin, Daniel D. Johnson, Jonathan Ho, Daniel Tarlow, and Rianne van den Berg. 2021. Structured denoising diffusion models in discrete state-spaces. In *Advances in Neural Information Processing Systems*.
- Paul Erdős and Alfréd Rényi. 1959. On random graphs I. *Publicationes Mathematicae Debrecen*, 6:290–297.
- Arthur Gretton, Karsten M. Borgwardt, Malte J. Rasch, Bernhard Schölkopf, and Alexander Smola. 2012. A kernel two-sample test. *Journal of Machine Learning Research*, 13:723–773.
- Aric A. Hagberg, Daniel A. Schult, and Pieter J. Swart. 2008. Exploring network structure, dynamics, and function using NetworkX. In *Proceedings of the 7th Python in Science Conference*, pages 11–15.
- Jonathan Ho, Ajay Jain, and Pieter Abbeel. 2020. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems*, volume 33, pages 6840–6851.
- Jonathan Ho and Tim Salimans. 2022. Classifier-free diffusion guidance. *arXiv preprint arXiv:2207.12598*.
- Tomaž Hočevar and Janez Demšar. 2014. A combinatorial approach to graphlet counting. *Bioinformatics*, 30(4):559–565.
- Jaehyeong Jo, Seul Lee, and Sung Ju Hwang. 2022. Score-based generative modeling of graphs via the

- system of stochastic differential equations. In *Proceedings of the 39th International Conference on Machine Learning*, pages 10362–10383.
- Renjie Liao, Yujia Li, Yang Song, Shenlong Wang, William Hamilton, David K. Duvenaud, Raquel Urtasun, and Richard Zemel. 2019. Efficient graph generation with graph recurrent attention networks. In *Advances in Neural Information Processing Systems*, pages 4255–4265.
- Derek Lim, Joshua Robinson, Lingxiao Zhao, Tess E. Smidt, Suvrit Sra, Haggai Maron, and Stefanie Jegelka. 2023. Sign and basis invariant networks for spectral graph representation learning. In *International Conference on Learning Representations*.
- Karolis Martinkus, Andreas Loukas, Nathanaël Peraudin, and Roger Wattenhofer. 2022. SPECTRE: Spectral conditioning helps to overcome the expressivity limits of one-shot graph generators. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*.
- Giorgia Minello, Alessandro Bicciato, Luca Rossi, Andrea Torsello, and Luca Cosmo. 2025. Generating graphs via spectral diffusion. In *The Thirteenth International Conference on Learning Representations*.
- Alexander Quinn Nichol and Prafulla Dhariwal. 2021. Improved denoising diffusion probabilistic models. In *Proceedings of the 38th International Conference on Machine Learning*, pages 8162–8171.
- Chenhao Niu, Yang Song, Jiaming Song, Shengjia Zhao, Aditya Grover, and Stefano Ermon. 2020. Permutation invariant graph generation via score-based generative modeling. In *Proceedings of the 23rd International Conference on Artificial Intelligence and Statistics*, volume 108, pages 4474–4484.
- Clement Vignac, Igor Krawczuk, Antoine Siraudin, Bohan Wang, Volkan Cevher, and Pascal Frossard. 2023. DiGress: Discrete denoising diffusion for graph generation. In *International Conference on Learning Representations*.
- Yuhang Xie and Sinno Jialin Pan. 2026. [Global and local topology-aware graph generation via dual conditioning diffusion](#). In *International Conference on Learning Representations*.
- Jiaxuan You, Rex Ying, Xiang Ren, William L. Hamilton, and Jure Leskovec. 2018. GraphRNN: Generating realistic graphs with deep auto-regressive models. In *Proceedings of the 35th International Conference on Machine Learning*.
- Cai Zhou, Xiyuan Wang, and Muhan Zhang. 2024. Unifying generation and prediction on graphs with latent graph diffusion. In *Advances in Neural Information Processing Systems*.
- Junyou Zhu, Langzhou He, Chao Gao, Dongpeng Hou, Zhen Su, Philip S. Yu, Jürgen Kurths, and Frank Hellmann. 2025. SDMG: Smoothing your diffusion models for powerful graph representation learning. In *Proceedings of the 42nd International Conference on Machine Learning*.